NIHR | National Institute for Health and Care Research

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| Reference number | 41627 |
| Programme | Research for Patient Benefit |
| Call | RfPB Competition 53 Stage 1 |
| Lead Applicant | Professor Nikos Donos |
| Host organisation | Barts Health NHS Trust |
| Research Title | Developing and refining strategies for improving implementation of supportive periodontal care in primary dental care |
| Start Date | 03/03/2025 |
| Grant Duration (Months) | 18 |
| End Date | |
| Estimated Research Cost | £200,000.00 |
| Estimated NHS Support & Treatment costs | £660.00 |
| Research Type | Primary Research |

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## Lead Applicant Details

**Lead Applicant**

| **Full Name** | Professor Nikos Donos | **Position** | Director of Research; Professor & Chair Periodontology & Implant Dentistry; HCC |
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| **ORCID iD** | 0000-0002-4117-9073 | **Institution** | Queen Mary University of London |

| **Specify your (lead applicant) role in the research** |
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| Professor Donos is an expert in clinical research and has been a reviewer for the European Clinical Guidelines (and adoption in UK) for the treatment of periodontal diseases including supportive periodontal care (SPC). He has completed an audit for 20 primary care practices that regularly refer patients to Royal London Dental Hospital allowing him to conceptualize the project. He has organized the team, designed the methods and will be responsible for the management and delivery of the different components of the study. He will be the lead for PPI and will have input in data analysis and interpretation. |

| **% FTE commitment** | 10 |
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## Research Team Details

| **Co-Applicant** | **Organisation** | **Specify Co-Applicant role in research** | **% FTE** |
|---|---|---|---|
| Dr Jamie Ross | Queen Mary University of London | Dr Ross is Senior Lecturer in Primary Care Sciences. She will contribute her expertise in implementation sciences, health psychology & PPI in medical research. | 5 |
| Dr Nikolaos Gkranias | Queen Mary University of London | Dr Gkranias has extensive experience in setting up, recruitment and delivering clinical research. Contributed in the design of the study and will participate in analysis. | 5 |
| Professor Eduardo Bernabe | Queen Mary University of London | Professor Bernabe has provided his expertise on recruitment of primary care dental practices and mixed-methods. He will supervise data analysis for the project. | 5 |
| Dr Elena Calciolari | Queen Mary University of London | Dr Calciolari has run focus groups that informed this project and will assist in the recruitment of practices at the Barts Health site. | 5 |
| Professor Nicholas Barker | University of Essex | Professor Barker is Deputy Chief Dental Officer and chair of East of England Local Dental Network and will lead PPI and primary care engagement. | 5 |

| **COI declaration**<br>Please declare any conflicts or potential conflicts of interest that you or your co-applicants may have in undertaking this research, including any relevant personal, non-personal and commercial interest that could be perceived as a conflict of interest. |
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| N/A |

| **Agreement to terms and conditions.** | ☒Confirmed |
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## Plain English Summary of Research

| Plain English summary of Reserarch |
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| **Background:** Gum disease (periodontitis) is a very common, chronic inflammation affecting the gums and the jawbone. When periodontitis is not treated, it may cause tooth loss, thus affecting speech, nutrition and overall wellbeing. Following periodontitis treatment in a specialist centre, patients are followed up at NHS primary care regularly over lifetime as part of a “supportive |

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periodontal care" (SPC) plan provided by the general dentist and dental team (hygienists and nurses). SPC is essential to save teeth, as relapse of gum disease may happen soon after completion of specialist treatment. However, the content and frequency of SPC is not well regulated within the NHS, resulting in gum disease relapse.

**Aims:** The purpose of this study is to (1) find barriers and enablers to SPC programmes in primary care dental practices, and to (2) create a strategy to address perceived barriers and maximise the potential enablers for SPC implementation.

**Methods:** We will recruit 4 dental practices from each of two recruitment sites (East London and East of England). In each practice, members of the clinical team (dentists, hygienists and receptionist/manager) and patients with gum disease will be invited to participate. The proposal will be completed in two stages. In the first stage, patients and the clinical team will be interviewed to identify barriers to implement SPC in primary care. The barriers will be matched to a set of potential solutions. In the second stage, the list of potential solutions will be ranked in importance by patients and the clinical team using an online questionnaire. This process will yield the solution that is most likely to succeed.

**Patient and public involvement:** A focus group of periodontal patients felt that SPC in primary care was not of appropriate frequency nor motivating. The project was seen as value for money and was suggested that the NHS should invest more in oral health as it affects wellbeing. The patients will help designing the final protocol, contribute to study meetings and will be involved in the recruitment of participants, interviews, data analysis and dissemination of the results.

**Dissemination:** Reporting in journals, conferences, webpages and via social media, podcasts and patients' forum.

## Research Plan

### Research Plan

**1) What is the problem being addressed?**

Periodontitis is an irreversible disease leading to tooth loss and it may reappear shortly after treatment without adequate supportive periodontal care (SPC) [1]. Hence, current national [2] and international treatment guidelines [3] highlight the importance of life-long SPC, which follows completion of periodontal therapy. SPC is essential for maintaining periodontal health, intercept disease relapse, and minimize tooth loss and associated masticatory complications [2, 3]. While active treatment (step 1, 2 and 3 of therapy) takes place at specialist secondary care (or by L2 accredited clinicians when secondary care is not possible), it is still expected that general dental practitioners (GDPs) in primary care take over life-long SPC (step 4 of therapy).

Our research group has shown that SPC at dental practices can be suboptimal or absent, leading to disease relapse and tooth loss in severe periodontitis patients [4]. Moreover, from anecdotal communication, it appears that only a small percentage of practices are currently implementing the staged course of treatment based on Business Service Authority (BSA) informal feedback.

**2) Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients and health care services?**

This project is in line with 3 of the top-10 oral health priorities by the James Lind Alliance Oral and Dental Health Priority Setting Partnership, that include improving access to dental services for the general public, finding the best way to prevent gum disease and reducing oral health inequalities at a community or population level [5].

Moreover, it will provide unique data to inform on the current barriers that prevent the correct implementation of SPC international standards to NHS primary care practices. Based on the identified barriers, we will be able to develop a specific intervention to successfully tackle them, thus creating the basis for a future trial (HTA grant) with the aim to improve dental healthcare services and limit/intercept periodontitis relapse, thus increasing the chances for tooth retain overtime, with positive consequences on oral and systemic health as well as patients' quality of life.

Based on the outcomes of our preliminary PPI work, it is likely that time and lack of training of the primary care dental team will be key perceived barriers. As such, one way to address those barriers (which we will elaborate as part of this project) will be to apply the philosophy of skill mix, which allows a more efficient management of the dental practice resources. Currently, all NHS courses of treatment in primary care have a GDP responsible to start them through examination, diagnosis, and treatment planning. Clinical reality shows that GDPs do not have the possibility of sharing responsibilities with other dental team members (e.g. hygienist) due to the lack of workforce and time, therefore they often end up completing all SPC-related tasks [6, 7]. However, following contract prototypes embracing use of wider skill mix, the concept of flexible commissioning was introduced (2019), allowing local variations in primary care contracts to better meet patients' needs. This project is in line with this new approach and NHS England can use it to validate flexible commissioning for SPC [7]. The use of a team approach to deliver the motivation and prevention phase, will free up time for the GDP to undertake more complex care, which will address the growing clinical time requirements for middle–aged patients [8, 9] who are more likely to require periodontal care. Ultimately, it will also reduce the NHS costs/burden associated with the inevitable vicious circle of re-referrals of relapsed patients to secondary care. Indeed, in January 2023, NHS England clarified that courses of treatment can be opened by hygienists and therapists, thus dispelling the previous belief that this was only within the purview of a dentist [7].

**3) Review of existing evidence – How does the existing literature support this proposal?**

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Severe periodontal disease is widespread, with a global prevalence of about 19% in people >15 years, representing more than 1 billion cases worldwide [10]. In the United Kingdom 70% of adults suffer from periodontitis, of which 10% present with severe forms [11]. It disproportionately affects the vulnerable segments of the population [12] and negatively impacts the quality of life [10], with a financial cost of $54 and $25 billion in direct and indirect treatment costs, respectively [10].

Periodontitis is an irreversible disease leading to tooth loss and may reappear shortly after treatment without adequate SPC (step 4 of therapy) [1]. According to current treatment guidelines [2, 3], SPC visits should last approximately 60 minutes and should include: medical history, re-evaluation (10-15 min); motivation, re-instruction and instrumentation (30-40 min); polishing, and determination of future recalls (8 min) [13]. No data are available on the adhesion/awareness to validated SPC within NHS or on the incidence of disease reoccurrence. This has resulted so far in a wide variation of SPC care delivered in terms of frequency, length and procedures.

Recently, an NHS document [14] in conjunction with the British Society of Periodontology guidance on delivering phased care for periodontitis patients under Units of Dental Activity (UDA) banding clarified the possibility of providing phased treatments in periodontitis patients. In particular, it was recognized that time is required to undertake SPC and, as such, a Band-2 course of treatment is permitted for engaging patients. During SPC dentists should repeat plaque scores, reinforce behaviour change, patient motivation, risk factor control, perform a detailed pocket charting, professional mechanical plaque removal and localised subgingival instrumentation of residual pockets. While this document is important as it recognises for the first time the importance of SPC in the NHS and provides some guidance on how to provide a staged treatment, it is not clear how it has been perceived and applied (if at all) by dentists, how much time they dedicate to SPC visits and how they decide on recall frequency.

No study has yet investigated the barriers to the correct implementation of SPC in the NHS. We know that an important part of SPC is motivation to achieve good oral hygiene, but there is lack of evidence for this in primary care, so this could potentially represent a barrier and it is often attributed to time constraints or an NHS contract that does not support prevention [15]. A systematic review concluded that interventions targeting goal setting, planning of behaviour and monitoring by a health care professional/patient, are effective in promoting behaviour change [16]. A meta-analysis on improving health messaging in clinical practice also supported the use of a tailored face-to-face cognitive psychological behavioural model to plan successful interventions in primary health care for increasing sustainability of the behaviour [17]. Another potentially critical factor is the knowledge of the dentist. SPC visits under specialist care in higher socioeconomic groups lead to better periodontal stability than SPC delivered by a GDP [18, 19]. However, life-long SPC within NHS secondary care is not applicable nor affordable, as access to specialists is limited, ultimately leading to social inequalities. Remarkably, when GDPs are instructed with a clear prescription of SPC tasks, this difference tends to disappear [20], so it is plausible that the lack of proper knowledge/training might be another barrier.

Conversely, adequate visit length and an efficient distribution of tasks within the dental team based on the use of skill mix in the NHS (which includes hygienists and extended duties dental nurses) are likely to improve SPC management in primary care. Dental team working, where skill mix is widely used, has been shown to be beneficial for individual clinicians, teams, practice owners and patients [21]. Within primary dental care, there is both research and practice-based evidence about the clinical effectiveness, safety, and acceptability of using dental therapists and hygienists to provide preventative and diagnostic care to patients, alongside performing a range of treatments to both adults and children [7]. However, it is still unclear how this approach is currently implemented for SPC management in the NHS.

**4) What is the research question / aims and objectives?**

This study forms an important initial step of a broader future study aimed at implementing and evaluating an intervention to improve the uptake of SPC in primary care (Figure). The current study aims to:

1) Identify contextual barriers and enablers related to the implementation of SPC programmes in primary care dental practices (Figure, stage 1).

2) Co-design a theoretically based implementation strategy that addresses perceived barriers and maximises the potential enablers of implementation (Figure, stage 2).

Further stages of the full implementation study will evaluate the feasibility of delivering the ad-hoc implementation strategy in a limited number of pilot dental practices and a full-scale implementation trial to evaluate whether the implementation strategy improves the uptake of SPC in primary care dental practices.

**5) Project plan**

**5.1. Study design**

This is a participatory, mixed-methods, implementation study, underpinned by the Consolidated Framework for Implementation Research (CFIR) [22]. The study will be carried out in two stages over a 18-month period (Figure 2). Ethical approval will be sought from the MHRA before the grant begins.

**5.2. Setting and participants**

The study will take place in general dental practices that have referred periodontal patients for specialised (secondary/tertiary) care to Royal Dental Hospital/Barts Health (QMUL) and secondary care dental services (University of Essex) during the past 5

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years. These specialised centres serve large, deprived and diverse populations in East London and East of England, respectively.

Following the completion of an audit at the Periodontology Department of Royal London Dental Hospital/Barts Health, we have already identified 20 primary care practices that regularly refer periodontal patients, while in East of England there is already an established Local Dental Network (LDN) of 780 dental practices with a reasonable amount of these patients requiring periodontal therapy (NB – co-applicant - is the chair of this LDN). In each of the two sites, we will recruit 4 practices to create an initial network. We will arrange visits to these primary care dental practices to discuss the project, check eligibility and invite them to participate. To be eligible, they must have a cohort of at least 3000 patients who regularly attend the practice and have a dental hygienist as part of the clinical team (i.e. they provide routine scaling and polishing in primary care) and provide SPC as part of routine care. We will recruit a diverse sample of participants from four key informant groups, namely dentists, dental hygienists, receptionists and patients, to ensure a thorough understanding of the implementation context and identify any potential implementation issues. Furthermore, collecting information from each informant group across different practices will allow separating local and cross-practice implementation issues. At least one dentist, one hygienist and one receptionist from each GDP, who must be involved in the administrative (i.e. referral process, booking appointments, etc) and/or clinical management of patients with periodontal disease will be recruited. We will also recruit up to 3 adult periodontal patients from each GDP who are at different stages of their SPC (i.e. those who are about to start SPC, those who are under SPC and those who never started SPC). The clinical team in each GDP will approach potential participants from those found eligible through dental records.

This initial sample size of 24 clinical team members (8 dentists, 8 hygienist and 8 receptionists) and 24 patients will provide sufficient information power [23] because the study aim is narrow and supported by an established theory (CFIR), the combination of participants is highly specific for the study aim, the interview dialogue is strong (experienced research assistant) and the analysis will be underpinned by a pre-determined framework.

**5.3. Stage 1: Understanding implementation barriers and enablers**

**5.3.1. Data collection**

The research assistant will conduct semi-structured interviews with key informant groups using a topic guide based on CFIR [22]. The updated CFIR was chosen because it breaks down contextual factors along several constructs organised in 5 domains relevant to healthcare interventions, namely innovation characteristics (8 constructs), outer setting (7), inner setting (11), implementation process (9), and patient individuals' characteristics (13) [24]. Therefore, it provides a systematic approach to identify factors that can encourage or impede the implementation of SPC in primary care. Interviews will be conducted face-to-face in a quiet and private room in each GDP. Every interview, lasting 30-45 min, will be audio-recorded, professionally transcribed and de-identified.

**5.3.2. Data analysis**

Interviews will be analysed using deductive and inductive approaches. A deductive coding strategy will be first used wherein implementation barriers and enablers that are identified during the coding process will be organised according to CFIR domains. In a subsequent stage, thematic content analysis via the framework method will be used to inductively develop categories from the raw data that reflect factors affecting implementation to cover the possibility that they were not covered in CFIR. This inductive approach will permit systematic and visible stages to the data analysis process, moving through five stages (familiarisation, identification, indexing, charting, mapping and interpretation) [25]. Combining both approaches will help examine how CFIR explains the findings and where it does not (thus allowing for theoretical contributions to the field), and can support in providing actionable, meaningful implications and recommendations locally.

Double-blind coding will be undertaken by the researcher assistant and a senior member of the research team (qualitative expert) independently at the start of the analysis process, who will then meet to refine the analytical framework, resolve disagreement and agree the final themes. The software NVivo (Lumivero, USA) will be used to manage data and to facilitate coding. Quality guidelines for the conduct of qualitative research will be adhered to [26].

**5.4. Stage 2: Co-design of the implementation strategy**

**5.4.1. Identifying implementation strategies**

The list of barriers and enablers identified in Stage 1 will be mapped to the implementation strategies that best address them using the CFIR-Expert Recommendations for Implementing Change (ERIC) matching tool [27, 28]. The ERIC is a compilation of 73 implementation strategies to mitigate or overcome anticipated barriers, which was developed by a panel of implementation science and clinical practice experts. The CFIR-ERIC matching tool links these two key resources to help users understand which ERIC implementation strategies will best address which CFIR barriers. The tool allows users to select the identified CFIR barriers; it then generates a prioritised list of high yield, targeted and relevant ERIC strategies that match each barrier. The list provides Level 1 endorsed ERIC strategies (>50% of experts ranked this as one of their top seven strategies for that barrier) and Level 2 endorsed ERIC strategies (20% to 50% of experts ranked this as one of their top seven strategies for that barrier). Users can thereafter select those strategies that best help to address each barrier and are most applicable to their context [27, 28]. To ensure consistency, two members of the research team will work together to map the selected strategies.

**5.4.2. Data collection**

A modified e-Delphi technique, including a combination of email invitations and two rounds of online surveys developed using

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Qualtrics®, will be used to reach consensus between the included participants on which implementation strategy to prioritise, as they might have different perceptions on barriers and their potential solutions. Using a consensus technique will ensure co-production of the implementation strategy, uniformity across included sites and teams, and engagement with later stages of the project. Participation will be anonymous to enable participants to comment freely and openly.

The list of recommended implementation strategies will be used to develop an online questionnaire, including relevant examples. This online questionnaire will be sent to all participating GDP practices, asking clinical teams and periodontal patients to indicate how important and feasible they believe each strategy is for SPC to be successfully implemented in primary care. Responses will be captured using 9-point ordinal scales (1 =least feasible/important, 9=most feasible/important). Reminders will be sent one and two weeks after the initial email. Participants' responses will be collated and summarised quantitatively. In round 2, a revised list of items will be circulated for ranking after analysis and discussion by the research team. Consensus will be achieved when at least 51% of the panellists agree with the implementation strategy in terms of both importance and feasibility. To mitigate barriers to consensus, we will also report areas of residual non-consensus and describe the nature of the dissent.

**5.4.3. Data analysis**

Descriptive quantitative analysis (response rate, median and interquartile range for each item) will allow ranking responses and identifying priorities. For each implementation strategy, we will calculate the median rating for both feasibility and importance. A total score will be calculated by adding together the median ratings of feasibility and importance for each implementation strategy. Then, strategies will be ranked based on their total score.

**6) Research Team**

This is a multidisciplinary team with expertise in primary and secondary care in dentistry and public health. The collaboration between secondary/primary care centres ensures adequate patient recruitment and representation of East London and East of England population in terms of socioeconomic and ethnic background. It also includes specialists in Public Health, Behavioural and Implementation Sciences and Pedagogy/Education. Research Support Service (RSS) was involved in study design, sample size calculation and will participate in data analysis.

**7) Timetable**

The proposal will be completed over 18 months (Timetable). Stage 1 will be completed from months 1 to 9, with two key deliverables: the recruitment of GDPs (clinical teams) and patients by month 6 and the mapping of identified barriers and facilitators to the CFIR framework by month 9. Stage 2 will be completed from months 10 to 18, including identifying the potential implementation strategies and reaching consensus on the best strategy to address the identified barriers by the end of the grant.

**8) PPI**

Two focus groups involving patients previously treated for periodontitis were proactively involved in providing feedback to the proposal, with a special focus on topic priority, recruitment strategies and editing the lay summary. The PPI group will attend the study steering group meetings and be involved in recruitment (producing adverts and patient information sheet and as point of contact for potential participants), data collection (carrying out interviews), data analysis (participants' de-briefing) and dissemination (creating reports and lay summaries).

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## Administrative contact details

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## Research and Development office contact details

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| R&D contact email address | c.g.colechin@qmul.ac.uk |
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## Uploads


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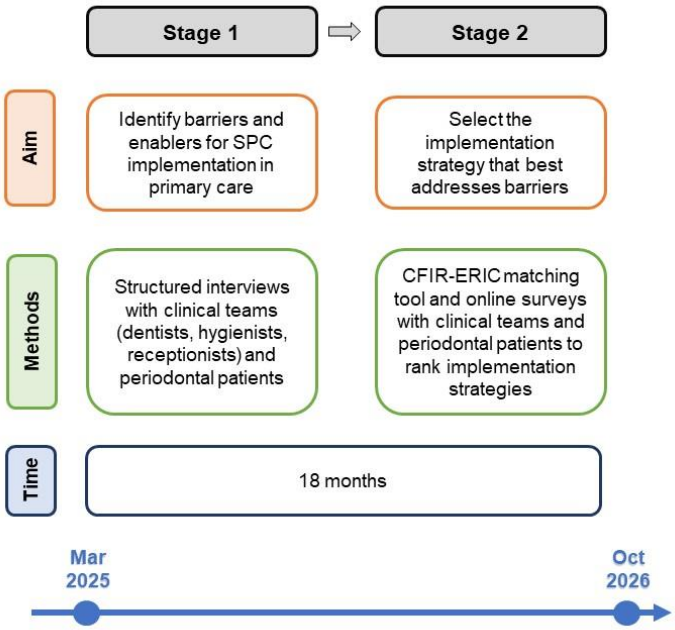
**Timetable.** Description of key activities and their timing during the 18-month grant period

| | | Pre-award | 2025 | | | | | | | | | 2026 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | *Mar* | *Apr* | *May* | *Jun* | *Jul* | *Aug* | *Sep* | *Oct* | *Nov* | *Dec* | *Jan* | *Feb* | *Mar* | *Apr* | *May* | *Jun* | *Jul* | *Aug* |
| ***Grant duration (months)*** | | | *1* | *2* | *3* | *4* | *5* | *6* | *7* | *8* | *9* | *10* | *11* | *12* | *13* | *14* | *15* | *16* | *17* | *18* |
| MHRA ethics approval | | | | | | | | | | | | | | | | | | | | |
| Recruitment of research support staff | | | | | | | | | | | | | | | | | | | | |
| Stage 1 | Recruitment of general dental practices | | | | | | | | | | | | | | | | | | | |
| Stage 1 | Recruitment of clinical team | | | | | | | | | | | | | | | | | | | |
| Stage 1 | Recruitment of patients | | | | | | | | | | | | | | | | | | | |
| Stage 1 | Interviews with clinical team | | | | | | | | | | | | | | | | | | | |
| Stage 1 | Interviews with patients | | | | | | | | | | | | | | | | | | | |
| Stage 1 | Qualitative data analysis (objective 1) | | | | | | | | | | | | | | | | | | | |
| Stage 2 | Matching barriers to strategies (ERIC-CFIR) | | | | | | | | | | | | | | | | | | | |
| Stage 2 | Delphi panel – round 1 | | | | | | | | | | | | | | | | | | | |
| Stage 2 | Data analysis of Delphi round 1 | | | | | | | | | | | | | | | | | | | |
| Stage 2 | Delphi panel – round 2 | | | | | | | | | | | | | | | | | | | |
| Stage 2 | Final data analysis round 2 (objective 2) | | | | | | | | | | | | | | | | | | | |
| Study Steering Group (SEG) meetings | | | | | | | | | | | | | | | | | | | | |
| Lay Advisory Group (LAG) meetings | | | | | | | | | | | | | | | | | | | | |
| Drafting and submission of manuscripts | | | | | | | | | | | | | | | | | | | | |
| Conference presentation | | | | | | | | | | | | | | | | | | | | |
| Dissemination activities | | | | | | | | | | | | | | | | | | | | |
| NIHR progress reports | | | | | | | | | | | | | | | | | | | | |

Figure. Design and timeline of the full implementation study.

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